Faculty of Sciences

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Master thesis

Procedural Generation of Solvable Cooperative Levels for the Laser Learning Environment

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Chapter 1

Introduction

1.1 Context

Reinforcement Learning (RL) has established itself as a powerful paradigm for training autonomous agents to make sequential decisions by interacting with an environment. In single-agent settings, an agent repeatedly observes the state of the world, selects an action, and receives a scalar reward signal that it tries to maximise over time. This framework has produced remarkable results in domains ranging from board games to robotic control.

Multi-Agent Reinforcement Learning (MARL) extends this paradigm to settings in which several agents act simultaneously within a shared environment. The agents may compete, cooperate, or coexist independently, and their policies influence one another's learning dynamics. Cooperative MARL, in particular, focuses on scenarios where a group of agents must jointly achieve a shared goal - a setting that arises naturally in applications such as robot swarm coordination, multi-robot task allocation, and cooperative strategy games.

Cooperative tasks introduce challenges that go beyond single-agent RL. Agents must not only explore a potentially vast joint action space, but they must do so while accounting for the behaviour of their teammates. Reward signals are often sparse: in many cooperative settings, the team receives a reward only upon completing the full task, leaving agents with no intermediate feedback to guide exploration. This makes the discovery of coordinated strategies especially difficult when the required joint behaviour is long and precisely ordered.

A dimension of cooperative MARL that is frequently underestimated is the role of the training environment itself. The structure of the levels or scenarios in which agents are trained directly determines what coordination behaviours they can discover and practise. A poorly designed environment - one that is unsolvable, trivially solvable by a single agent, or repetitive in structure - yields limited training signal and constrains generalisation. Conversely, a well-designed set of environments can expose agents to a rich variety of coordination challenges, accelerate learning, and produce more robust policies. Designing such environments at scale, however, is a labour-intensive process when done manually.

1.2 Motivation

Procedural Content Generation (PCG) offers an alternative: the algorithmic, automatic creation of environments according to user-defined criteria. Originally developed for video game level design, PCG has since been applied to simulation-based agent training, where it enables the production of large, diverse sets of environments without manual intervention. In the MARL context, PCG can supply a steady stream of novel levels during training, reducing the risk of overfitting to a fixed set of hand-crafted scenarios and supporting generalisation across varied task structures.

The central difficulty in applying PCG to cooperative MARL is ensuring that generated environments are meaningful: not only structurally well-formed, but genuinely solvable, and genuinely cooperative. These are non-trivial requirements.

Solvability in a multi-agent setting is more demanding than in single-agent settings. It is not sufficient for each agent to have a path to its goal; the agents must be able to reach their goals simultaneously, following a joint sequence of actions that is consistent with all environmental constraints - including dynamic obstacles such as laser beams whose activity depends on the current positions of other agents. Testing whether such a joint solution exists is computationally non-trivial, and naively generating random environments yields mostly unsolvable configurations, especially as the grid size and number of agents grow.

Cooperation must be structurally enforced. A level that can be solved by agents acting independently - without any agent ever assisting another - provides no training signal for cooperative behaviour. For a level to serve as a useful cooperative training instance, it must have the property that no agent can reach its goal unless at least one other agent performs a supporting action. Simply generating solvable levels does not guarantee this: the generator must be designed or constrained to produce configurations in which cooperation is a structural necessity, not an option.

This thesis addresses both requirements together. We focus on the Laser Learning Environment (LLE) [1], a 2D grid-based cooperative benchmark for MARL, as our primary instantiation. LLE is particularly well-suited to this study: its laser-blocking mechanics create precisely the kind of inter-agent dependencies that make cooperation structurally necessary in certain configurations, and its well-defined grid representation is amenable to formal analysis.

1.3 Problem Statement

The central problem this thesis addresses is the following: how can we automatically generate LLE levels that are provably solvable and provably require inter-agent cooperation?

We target two formal properties and one broader design objective:

1. **Solvability** - the level admits at least one valid joint action sequence through which the agents collectively occupy all exit tiles. This is a necessary baseline: a level that cannot be completed is useless for training.
2. **Cooperation requirement** - the level cannot be solved without at least one agent performing a cooperative act (specifically, blocking a laser of its own colour to allow a teammate to pass). This property ensures that the level is not trivially solvable by independent agent behaviour.
3. **Learnability** - the generated instances should expose coordination patterns that are plausible training signals for MARL algorithms. Unlike solvability and cooperation, learnability is not

formalised or certified in this thesis; we treat it as a design objective and discuss it in terms of structural affordances that may facilitate exploration.

The approach we adopt to address the first two properties is a reduction to Boolean Satisfiability (SAT): we encode the constraints of the LLE level as a propositional formula in conjunctive normal form (CNF), and delegate the solvability check to a modern SAT solver. The cooperation property is detected via a second SAT call on a stricter variant of the encoding. Both calls are embedded in a generation loop that produces levels until the desired properties are satisfied.

1.4 Contributions

This thesis makes the following contributions:

- **A SAT-based solver for bounded-horizon LLE solvability.** We provide a CNF encoding of the LLE decision problem over a bounded time horizon T. Given a level and a horizon, the solver returns SAT (with a satisfying assignment that encodes a valid joint trajectory) or UNSAT (certifying that no solution exists within T steps). The encoding is described in full in .
- **A formal cooperation detector.** We define a strict variant of the LLE semantics in which agents can no longer block beams of their own colour, thereby removing the laser-blocking action through which agents help one another. We prove that a level requires cooperation if and only if the standard encoding is satisfiable and the strict encoding is unsatisfiable. This gives a decision procedure for the cooperation property based on two SAT calls ().
- **A family of procedural level generators.** Building on the solver and cooperation detector, we implement six generators: random solvable, constrained random solvable, random cooperative, constrained random cooperative, constructive solvable, and constructive cooperative. Each accepted level is certified by the solver to satisfy the advertised properties of its output ().
- **An empirical comparison of two SAT formulations.** We compare two alternative encodings of the agent-uniqueness constraint on four benchmark levels, measuring their effect on CNF size, model generation time, and solver runtime (,).

The approach developed here is specific to LLE in its SAT encoding - the laser mechanics, colour-matching rules, and exit conditions are all embedded in the formula structure. However, the broader methodology - reducing solvability and cooperation to formal properties, verifying them via a constraint solver, and embedding the verifier in a generation loop - is general and can be adapted to other grid-based cooperative MARL environments, provided an appropriate encoding is designed for the target environment.

1.5 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 - Background introduces the technical foundations needed to follow the thesis: the MARL framework and its cooperative variant, a detailed description of the Laser Learning Environment, an overview of Procedural Content Generation and its main paradigms, and an introduction to Boolean Satisfiability including its complexity-theoretic role in the context of LLE.

Chapter 3 - Related Work positions the thesis with respect to prior work on procedural content generation, solvability-aware generation, and cooperative MARL benchmarks.

Chapter 4 - Methods is the core of the thesis. It formalises the LLE level structure, solvability, and cooperation requirement; presents the full SAT encoding and its correctness argument; introduces the cooperation detection theorem and its proof; describes the six generators; and defines the benchmarking protocol.

Chapter 5 - Experiments presents the empirical results on solver performance, CNF size, and the effect of the two movement formulations on benchmark instances.

Chapter 6 - Conclusion summarises the main findings, discusses the limitations of the current approach, and outlines future extensions.

Chapter 2

Background

2.1 Multi-Agent Reinforcement Learning

2.1.1 Single-Agent Reinforcement Learning

Reinforcement Learning (RL) is a framework for sequential decision-making in which an agent interacts with an environment over a series of discrete time steps [2]. At each step t, the agent observes a state s_t from the state space X, selects an action u_t from its action space U, and receives a scalar reward $r_t \in \mathbb{R}$ from the environment. The environment then transitions to a new state s_{t+1} according to a transition function $T : X \times U \to \Delta(X)$, where $\Delta(X)$ denotes the set of probability distributions over X. This structure defines a Markov Decision Process (MDP), characterised by the tuple (X, U, T, R, γ), where $R : X \times U \to \mathbb{R}$ is the reward function and $\gamma \in [0, 1)$ is a discount factor controlling the relative weight of future rewards.

The agent's goal is to learn a policy $\pi : X \to \Delta(U)$ that maximises the expected discounted cumulative reward, or return:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$$

RL algorithms differ in how they estimate and optimise this objective - for example through value function approximation, policy-gradient methods, or actor-critic architectures - but all share the same underlying MDP structure [2].

2.1.2 Extension to Multiple Agents

Multi-Agent Reinforcement Learning (MARL) extends the single-agent framework to settings involving $n \geq 2$ agents acting simultaneously within a shared environment. A standard formalisation is the stochastic game introduced by Shapley [3], and widely used as a foundation for MARL [4]. The model is given by the tuple $\left(X, \{U^i\}_{i=1}^n, T, \{R^i\}_{i=1}^n, \gamma\right)$, where U^i is the action space of agent i and R^i is its reward function. At each time step, every agent selects an action independently, the joint action $\boldsymbol{u}_t = (u_t^1, ..., u_t^n)$ determines the state transition, and each agent receives a reward.

Depending on the structure of the reward functions, stochastic games span a spectrum from fully competitive (zero-sum games, where agents' interests are directly opposed) to fully cooperative (team games, where all agents share the same reward). Intermediate settings - with a mix of individual and shared incentives - also arise in practice. This thesis focuses exclusively on the cooperative setting.

In cooperative MARL, all agents share a single reward signal: $R^1 = R^2 = ... = R^n = R$. The team's objective is to find a joint policy $\boldsymbol{\pi} = (\pi^1, ..., \pi^n)$ that maximises the shared return. Cooperation is thus not assumed but must emerge from learning: agents must discover that coordinating their actions leads to higher team rewards than acting independently.

2.1.3 Challenges in Cooperative MARL

Cooperative MARL introduces a set of challenges that are absent or less severe in single-agent settings.

Non-stationarity. From the perspective of any individual agent, the environment appears non-stationary: other agents are simultaneously updating their policies, causing the effective dynamics observed by each agent to change over time. This violates the stationarity assumption underlying most single-agent RL theory and can destabilise learning.

Exponential joint action space. The joint action space grows exponentially in the number of agents. Even when each agent has a small individual action space, the number of possible joint actions becomes intractable for large teams. This makes centralised control infeasible and motivates decentralised execution paradigms.

Credit assignment. When agents share a single reward signal, attributing each agent's contribution to the team outcome becomes difficult. Poor credit assignment can lead to agents converging to suboptimal policies in which some agents free-ride on the efforts of others.

Sparse rewards and exploration. In many cooperative tasks, the team reward is issued only upon completing the full task. Agents must therefore explore long sequences of joint actions without intermediate feedback. In environments where the rewarding joint behaviour is rare or precisely ordered, pure random exploration is ineffective and training can stall entirely.

This last challenge is particularly acute in environments like LLE, where agents must execute carefully synchronised cooperative sequences with no intermediate reward - a setting known as the zero-incentive bottleneck problem, described in the following section.

2.2 The Laser Learning Environment

The Laser Learning Environment (LLE) [1] is a 2D grid-based cooperative puzzle game designed as a benchmark for MARL algorithms. It supports between one and four agents and is built around a set of mechanics that make cooperative coordination both necessary and difficult to learn.

2.2.1 Structure and Mechanics

An LLE level is played on a rectangular grid of height H and width W. Each cell is either a floor tile, a wall, a laser source, or an exit. The key components are as follows.

Agents. Each agent is assigned a unique colour from the set $C = \{0, 1, ..., n_a - 1\}$. At the start of an episode, agent c is placed at its designated starting cell $s(c)$. At each time step, an agent may stay in place or move to an adjacent non-wall cell. Agents cannot occupy the same cell simultaneously.

Lasers. Each laser source $(c, d, p) \in \mathcal{S}$ emits a beam of colour c in direction $d \in \{N, S, E, W\}$ starting from cell p. The beam propagates cell by cell in direction d until it is blocked by a wall or, crucially, by an agent of the same colour. Any agent of a different colour that enters a cell traversed by a beam of colour c is eliminated. This creates a strict interdependency: each laser can only be neutralised by one specific agent.

Exits. A level contains exactly n_a exit tiles. The episode is completed successfully when all exit tiles are occupied simultaneously. Since agents cannot share positions, this means that each exit is occupied by exactly one agent in a winning configuration.

Zero-incentive bottlenecks. A distinctive property of LLE is that intermediate cooperative acts - such as positioning to block a laser for a teammate - carry no immediate reward. The reward is issued only when all agents complete the level. An agent that sacrifices its own optimal path to assist a teammate receives no direct signal that this behaviour is beneficial. This creates exploration bottlenecks: agents have no local incentive to adopt cooperative strategies, and discovering the required joint behaviour through random exploration is exceedingly unlikely, especially in larger levels [1].

The full LLE implementation contains additional mechanics that are not modelled in this thesis, such as gems that provide rewards and void tiles that eliminate agents. We intentionally omit these mechanics from the formal model because they are not needed for the bounded solvability questions studied here. In particular, void tiles can be conservatively replaced by walls when constructing instances for the current solver. Likewise, when we move to the SAT model, laser contact is encoded as forbidden occupancy rather than as an explicit elimination transition. This simplification is sound for the bounded reachability questions studied here because any trajectory that steps onto a deadly beam is losing and therefore irrelevant to solvability.

2.2.2 An Illustrative Example

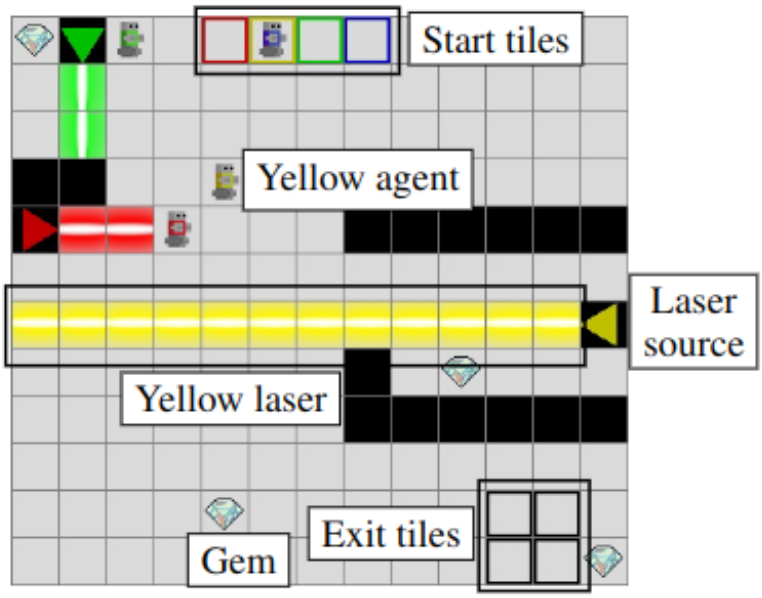

Figure 1: An annotated LLE level. Laser beams (coloured lines) block movement for agents of a different colour. The yellow agent blocks the yellow laser to allow the other agents to pass - a cooperative act that yields no intermediate reward.

The figure above illustrates a typical LLE level. The yellow agent must position itself in the path of the yellow laser, blocking it so that the other agents can pass through cells the beam would otherwise reach. This blocking act is purely beneficial to the other agents: the yellow agent is immune to its own laser and gains nothing directly from the manoeuvre. The level cannot be completed without this cooperative step, yet no reward is given for performing it.

2.2.3 Relevance as a Benchmark

LLE has been used to evaluate several state-of-the-art value-based MARL algorithms, including methods augmented with prioritised experience replay, n-step returns, and intrinsic curiosity via random network distillation [1]. Despite these enhancements, agents frequently fail to escape coordination bottlenecks in LLE, even when they achieve near-perfect coordination in more conventional cooperative benchmarks.

This combination of properties - formal solvability structure, strict coordination requirements, sparse rewards, and zero-incentive bottlenecks - makes LLE an ideal environment for studying the relationship between level design and agent learning. It is the primary instantiation of our framework, though the methodology developed here is designed with generalisability to other grid-based cooperative environments in mind.

2.3 Procedural Content Generation

Procedural Content Generation (PCG) refers to the algorithmic creation of content - levels, maps, textures, narratives, or entire game worlds - with limited or no direct human input. PCG has a long history in game development, where it serves to increase replayability, reduce production costs, and produce content at scales that manual design cannot achieve [5]. More recently, PCG has been

adopted in simulation-based agent training as a way to supply diverse and varied environments during learning [6].

PCG methods can be broadly organised into three paradigms.

Constructive methods generate content in a single forward pass, without backtracking or verification. Examples include noise-based generators that produce cave-like or organic layouts, grammar-based systems that expand structural rules into level geometry, and Wave Function Collapse, which tiles levels from locally consistent patterns. Constructive methods are fast and scalable, but offer no formal guarantees: a generated level may be unsolvable, disconnected, or trivially easy [5].

Search-based methods frame content generation as an optimisation problem and explore the space of possible levels using heuristic search or evolutionary algorithms. A fitness function evaluates candidate levels against desired properties - solvability, playability, difficulty - and the search is guided toward high-fitness regions. Search-based methods can incorporate complex objectives but are computationally expensive and depend heavily on the quality of the fitness function [6].

Constraint-based methods encode the desired properties of the content as a constraint satisfaction problem (CSP) or a propositional formula, and use a solver to find a level that satisfies all constraints. This approach offers formal guarantees: if a solution is found, it is certified to satisfy every specified property. The cost is the computational overhead of solving the constraint system, which grows with the complexity of the level.

This thesis adopts the constraint-based paradigm, using Boolean Satisfiability as the underlying solver. The choice is motivated by the need for formal solvability and cooperation guarantees - properties that constructive and search-based methods cannot provide directly.

2.4 Boolean Satisfiability

2.4.1 The SAT Problem

The Boolean Satisfiability problem (SAT) is the canonical NP-complete decision problem. Given a propositional formula φ over a set of Boolean variables $x_1, ..., x_n$, the question is: does there exist a truth assignment $\nu : \{x_1, ..., x_n\} \to \{\text{true}, \text{false}\}$ such that $\varphi(\nu) = \text{true}$? The central role of SAT in complexity theory was established by the Cook-Levin theorem [7].

SAT solvers operate on formulas in **Conjunctive Normal Form** (CNF): a conjunction of clauses, where each clause is a disjunction of literals (a variable or its negation):

$$\varphi = \bigwedge_{i=1}^{m} \bigvee_{j} \ell_{i,j}$$

Any propositional formula can be converted to an equisatisfiable CNF formula in polynomial time using standard transformations. When a formula is satisfiable, a solver returns a satisfying assignment (a **model**); when it is not, it returns UNSAT.

2.4.2 Modern SAT Solvers

Early SAT solvers were based on the Davis-Putnam-Logemann-Loveland (DPLL) algorithm, a backtracking search procedure augmented with unit propagation and related simplification rules [8]. Modern solvers implement Conflict-Driven Clause Learning (CDCL), which extends DPLL with two key innovations: **clause learning** (recording the reason for each conflict as a new clause, preventing the solver from revisiting the same conflict) and **non-chronological backtracking** (jumping back further in the search tree based on the learned clause). CDCL solvers also employ sophisticated heuristics for variable ordering and restart policies that dramatically reduce the search space in practice.

Despite SAT being NP-complete in the worst case, modern CDCL solvers routinely handle very large industrial instances efficiently in practice. This empirical tractability is the primary motivation for using SAT as the backend for our solvability checker.

2.4.3 Complexity of LLE Solvability

The solvability problem for LLE - given a level L and a time horizon T, does a valid joint trajectory of length at most T exist? - lies in **NP**: a candidate trajectory can be verified in polynomial time by simulating the joint execution and checking that all agents occupy the exit tiles at the end without violating any constraint.

The reduction we present in encodes this problem as a CNF formula in polynomial time with respect to the level parameters (H, W, $|C|$, T). This establishes that LLE solvability is polynomial-time many-one reducible to SAT:

$$\text{LLE-Solvability} \leq_p \text{SAT}$$

This means LLE solvability is **at most as hard as SAT**: any algorithm for SAT immediately yields an algorithm for LLE solvability with only polynomial overhead.

Whether LLE solvability is **NP-hard** - that is, whether SAT is also reducible to LLE solvability - remains an open question. Establishing NP-hardness would require constructing a polynomial-time reduction from a known NP-hard problem to LLE solvability, thereby showing that LLE solvability can encode arbitrary constraint-satisfaction structure. This thesis does not claim such a result.

It is also important to distinguish proved statements from standard complexity-theoretic beliefs. The question whether $\mathrm{P} = \mathrm{NP}$ remains open. Accordingly, statements of the form “no polynomial-time algorithm exists” should not be read here as proved impossibility results. What we can state formally is that SAT is NP-complete, that bounded-horizon LLE solvability belongs to NP, and that our reduction shows bounded-horizon LLE solvability is polynomial-time reducible to SAT.

The practical consequence of the reduction is nevertheless clear: our SAT-based solver inherits the strong empirical performance of modern SAT solvers, making it effective on the LLE instances encountered in practice even though the worst-case complexity remains exponential.

Chapter 3

Related Work

Chapter 4

Methods

4.1 Problem Formalization

We now formally define the objects and properties at the centre of this thesis. This section operates at the level of the game semantics; the SAT encoding of these properties follows in .

The LLE environment supports between 1 and 4 agents. Accordingly, throughout this thesis we assume that $1 \leq n_a \leq 4$.

4.1.1 LLE Level

An LLE level is a tuple $L = (H, W, C, s, \mathcal{W}, \mathcal{S}, \mathcal{E})$ where:

- $H, W \in \mathbb{N}^+$ are the height and width of the grid.
- $P = \{(x, y) \mid 0 \leq x < W, 0 \leq y < H\}$ is the set of all grid positions.
- $C = \{0, 1, ..., n_a - 1\}$ is the set of agent colours, with $1 \leq n_a \leq 4$.
- $s : C \to P$ assigns an initial position to each agent.
- $\mathcal{W} \subseteq P$ is the set of wall positions.
- $D = \{N, S, E, W\}$ is the set of laser directions.
- $\mathcal{S} \subseteq C \times D \times P$ is the set of laser sources. A source $(c, d, p) \in \mathcal{S}$ emits a laser of colour c in direction d from position p.
- In the instances studied in this thesis, each colour appears in at most one laser source.
- $\mathcal{E} \subseteq P$ is the set of exit positions, with $|\mathcal{E}| = |C|$.

4.1.2 Joint Trajectory

A **horizon** $T \in \mathbb{N}$ is the maximum number of joint moves allowed in a candidate solution.

A **joint trajectory** of length T is a sequence of joint positions $\sigma = (p_0, p_1, ..., p_T)$ where each $p_t : C \to P$ assigns a position to every agent at time step t.

A joint trajectory is **valid** if it satisfies the following conditions:

1. **Initialization:** $p_0(c) = s(c)$ for all $c \in C$.

2. **Movement:** For all $t < T$ and all $c \in C$, $p_{t+1}(c)$ is reachable from $p_{t(c)}$ in one step: the agent may stay in place or move to a 4-neighbouring cell, but it may not move into a wall or a laser-source cell.
3. **No collision:** $p_{t(c_1)} \neq p_{t(c_2)}$ for all t and all distinct $c_1, c_2 \in C$.
4. **Conservative hand-off rule:** For all $t < T$ and all distinct $c_1, c_2 \in C$, agents may not enter a cell occupied by another agent at the previous time step; that is, $p_{t+1}(c_1) \neq p_{t(c_2)}$ and $p_{t(c_1)} \neq p_{t+1}(c_2)$. This matches the collision model enforced by the SAT encoding used in the implementation.
5. **Laser safety:** No agent c_1 occupies a cell at time t where a laser of colour $c_2 \neq c_1$ is active.
6. **Stay on exit:** If $p_{t(c)} \in \mathcal{E}$ for some $t < T$, then $p_{t+1}(c) = p_{t(c)}$.

4.1.3 Solvability

A level L is **solvable** with horizon T if there exists a valid joint trajectory σ of length T such that the set of occupied positions at time T is exactly the set of exits:

$$\left\{p_{T(c)} \mid c \in C\right\} = \mathcal{E}$$

A level is **solvable** without qualification if it is solvable for some finite horizon.

The restriction to a bounded horizon is natural for the SAT encoding. In the LLE mechanics studied here, laser activity at time t is a deterministic function of the joint agent positions at time t. Consequently, the full game state is determined by the joint position map p_t. If a trajectory repeats the same joint position twice, the intervening segment forms a loop and can be removed without affecting reachability of later states. Therefore, if a level is solvable at all, it is solvable within a finite horizon bounded by the number of collision-free joint configurations.

4.1.4 Cooperation Requirement

Informally, a level **requires cooperation** with horizon T if no solution of length at most T exists unless at least one agent blocks a laser of its own colour to make progress possible for another agent.

We make this precise via **strict laser semantics**. In the strict variant, agents are no longer immune to lasers of their own colour. Every agent is blocked by every active laser, regardless of colour matching. A valid trajectory under strict semantics is called a **strict trajectory**.

This definition matches the current LLE mechanics because, in the model studied here, an agent can change the traversable space available to its teammates only by standing on a beam of its own colour. Strict semantics removes exactly this possibility while leaving the rest of the level dynamics unchanged.

Definition (Cooperation Requirement with horizon T). A level L is said to **require cooperation** with horizon T if:

1. L is solvable under the standard semantics, and
2. L admits no valid strict trajectory whose final positions occupy all exit tiles.

When the horizon is clear from context, we simply say that L requires cooperation. This bounded form is the one used throughout the rest of the methods chapter, since both the SAT solver and the cooperation detector operate at a fixed finite horizon.

The formal proof that this captures the intended notion of cooperation is given in .

4.2 Solver by Reduction to SAT

4.2.1 Definition of Sets

We reuse the level objects introduced in : the grid dimensions H and W, the position set P, the colour set C, the direction set D, the wall set $\mathcal{W}$, the source set $\mathcal{S}$, the exit set $\mathcal{E}$, and the start map s.

The SAT reduction introduces a finite horizon $T_{\max} \in \mathbb{N}$, which is the maximum number of joint moves allowed in the bounded decision problem, together with the discrete time-step set $T = \{0, 1, ..., T_{\max}\}$.

We also define

$$P_{\text{src}} = \{p \in P \mid \exists c \in C, d \in D : (c, d, p) \in \mathcal{S}\}$$

for the set of source positions.

To match the benchmark and generated instances studied in this thesis, we assume that each colour appears in at most one laser source. Under this assumption, when a source of colour c exists, its position and direction are uniquely determined by c.

We also write $\text{start}(c) \in P$ for the initial position of agent $c \in C$.

4.2.2 Definition of Variables

We introduce three families of propositional variables.

- $a_{c,x,y,t}$: true iff agent $c \in C$ occupies position $(x, y) \in P$ at time step $t \in T$.
- $b_{c,d,x,y,t}$: true iff there exists $p_s \in P$ such that $(c, d, p_s) \in \mathcal{S}$ and the beam emitted by that source is active at position $(x, y) \in P$ at time $t \in T$.
- $l_{c,x,y,t}$: true iff a laser of colour $c \in C$ is active at position $(x, y) \in P$ at time $t \in T$.

4.2.3 Constraints and Logical Encoding

We now formalise the constraint families used in the reduction. Each group of clauses corresponds to one logical component of the bounded-horizon decision problem.

1. **Initialization**
   - **Agents:**
     Each agent $c \in C$ is placed at its designated starting position $\text{start}(c)$ at $t = 0$; all other positions are unoccupied by that agent:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in P} \begin{cases} a_{c,x,y,0} & \text{if } (x, y) = \text{start}(c) \\ \neg a_{c,x,y,0} & \text{otherwise} \end{cases}$$

- **Laser sources:**
  For each laser source $(c, d, (x_s, y_s)) \in \mathcal{S}$, the beam is always active at its origin, at every time step:

$$\bigwedge_{(c,d,(x_s,y_s))\in\mathcal{S}} \bigwedge_{t\in T} b_{c,d,x_s,y_s,t}$$

2. **Agent Movements**

We define the set of positions reachable from (x, y) in one step as (x, y) itself together with its four grid neighbours, excluding walls and laser-source positions:

$$\begin{aligned}\text{next}(x, y) = \{(x', y') \in \{(x, y), (x, y-1), (x+1, y), (x, y+1), (x-1, y)\} \mid \\ (x', y') \in P, (x', y') \notin \mathcal{W}, (x', y') \notin P_{\text{src}}\}\end{aligned}$$

- **Forward consistency:**
  If agent c is at position (x, y) at time t, it must be at some position in $\text{next}(x, y)$ at time $t + 1$:

$$\bigwedge_{c\in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y)\in P} a_{c,x,y,t} \rightarrow \bigvee_{(x',y')\in \text{next}(x,y)} a_{c,x',y',t+1}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y)\in P} \neg a_{c,x,y,t} \vee \bigvee_{(x',y')\in \text{next}(x,y)} a_{c,x',y',t+1}$$

Two methods are proposed to enforce that each agent occupies at most one position at each time step.

- **Global uniqueness:**
  No two distinct positions can both be occupied by agent c at time t:

$$\bigwedge_{c\in C} \bigwedge_{t\in T} \bigwedge_{(x_1,y_1)\in P} \bigwedge_{\substack{(x_2,y_2)\in P,\\ (x_2,y_2)\neq(x_1,y_1)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

- **Local uniqueness:**
  No two distinct positions in $\text{next}(x, y)$ can simultaneously be occupied by agent c at time $t + 1$:

$$\bigwedge_{c\in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y)\in P} \bigwedge_{(x',y')\in \text{next}(x,y)} \bigwedge_{\substack{(x'',y'')\in \text{next}(x,y),\\ (x'',y'')\neq(x',y')}} \neg a_{c,x',y',t+1} \vee \neg a_{c,x'',y'',t+1}$$

Backward consistency:
If agent c is at position (x, y) at time $t + 1$, it must have been at some position in $\text{next}(x, y)$ at time t:

$$\bigwedge_{c\in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y)\in P} a_{c,x,y,t+1} \rightarrow \bigvee_{(x',y')\in \text{next}(x,y)} a_{c,x',y',t}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C}\bigwedge_{t=0}^{T_{\max}-1}\bigwedge_{(x,y)\in P}\neg a_{c,x,y,t+1}\vee\bigvee_{(x',y')\in\,\mathrm{next}(x,y)}a_{c,x',y',t}$$

- **No simultaneous occupation:**
  Two distinct agents cannot share the same position at the same time, nor can they move to a position already occupied by the other agent:

$$\bigwedge_{c_1\in C}\bigwedge_{c_2\in C,c_2\neq c_1}\bigwedge_{(x,y)\in P}\bigwedge_{t\in T}\neg a_{c_1,x,y,t}\vee\neg a_{c_2,x,y,t}$$

$$\bigwedge_{c_1\in C}\bigwedge_{c_2\in C,c_2\neq c_1}\bigwedge_{(x,y)\in P}\bigwedge_{t=0}^{T_{\max}-1}\neg a_{c_1,x,y,t+1}\vee\neg a_{c_2,x,y,t}$$

$$\bigwedge_{c_1\in C}\bigwedge_{c_2\in C,c_2\neq c_1}\bigwedge_{(x,y)\in P}\bigwedge_{t=0}^{T_{\max}-1}\neg a_{c_1,x,y,t}\vee\neg a_{c_2,x,y,t+1}$$

- **Victory condition:**
  Each exit must be occupied by at least one agent at time $T_{\max}$. Since $|\mathcal{E}|=n_a$ and agents cannot share positions, this enforces a bijection between agents and exits:

$$\bigwedge_{(x,y)\in\mathcal{E}}\bigvee_{c\in C}a_{c,x,y,T_{\max}}$$

- **Stay on exit:**
  Once an agent reaches an exit, it remains there for all subsequent time steps:

$$\bigwedge_{c\in C}\bigwedge_{(x,y)\in\mathcal{E}}\bigwedge_{t=0}^{T_{\max}-1}\neg a_{c,x,y,t}\vee a_{c,x,y,t+1}$$

3. **Laser Activity**

We define $\mathrm{next}_{d(x,y)}$ as the position immediately adjacent to (x,y) in direction d:

$$\mathrm{next}_{d(x,y)}=\begin{cases}(x,y-1)\text{ if }d=N\\(x+1,y)\text{ if }d=E\\(x,y+1)\text{ if }d=S\\(x-1,y)\text{ if }d=W\end{cases}$$

- **Walls block beams:**
  A beam cannot be active at a wall position:

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}}\bigwedge_{(x,y)\in\mathcal{W}}\bigwedge_{t\in T}\neg b_{c,d,x,y,t}$$

- **Beam propagation:**
  Beam propagation clauses are instantiated only when the successor cell $(x',y')=\mathrm{next}_{d(x,y)}$ lies inside the grid, is not a wall, and is not itself a source cell. Source cells are handled separately by the initialization clauses above. Under these conditions, the beam is active at (x',y') iff it is active at (x,y) and no agent of colour c occupies (x',y'):

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\notin P_{\text{src}}}} b_{c,d,x',y',t} \leftrightarrow \left(b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}\right)$$

$$\Updownarrow$$

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\notin P_{\text{src}}}} \neg b_{c,d,x,y,t} \vee a_{c,x',y',t} \vee b_{c,d,x',y',t}$$

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\notin P_{\text{src}}}} b_{c,d,x,y,t} \vee \neg b_{c,d,x',y',t}$$

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\in P\setminus\mathcal{W},\\ \text{next}_{d(x,y)}\notin P_{\text{src}}}} \neg a_{c,x',y',t} \vee \neg b_{c,d,x',y',t}$$

- **Link between beam and laser variables:**
  Since each colour has at most one source in the instances considered here, the laser variable $l_{c,x,y,t}$ is true at a position iff the beam of the unique source of colour c is active there:

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} b_{c,d,x,y,t} \leftrightarrow l_{c,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{(c,d,p_s)\in\mathcal{S}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \left(\neg b_{c,d,x,y,t} \vee l_{c,x,y,t}\right) \wedge \left(\neg l_{c,x,y,t} \vee b_{c,d,x,y,t}\right)$$

- **Agents cannot step on active lasers:**
  An agent of colour c_1 cannot occupy a position where a laser of colour $c_2 \neq c_1$ is active. Agents are immune only to lasers of their own colour:

$$\bigwedge_{c_1\in C} \bigwedge_{\substack{c_2\in C,\\ c_2\neq c_1}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} l_{c_2,x,y,t} \rightarrow \neg a_{c_1,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{c_1\in C} \bigwedge_{\substack{c_2\in C,\\ c_2\neq c_1}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg l_{c_2,x,y,t} \vee \neg a_{c_1,x,y,t}$$

4.2.4 Clause Complexity and Polynomial Size

To show that the reduction has polynomial size, it is enough to bound the number of generated clauses as a function of the input parameters. Let

$$
\begin{aligned}
n &= |C|, \\
p &= |P| = HW, \\
s &= |\mathcal{S}|, \\
e &= |\mathcal{E}|, \\
\tau &= T_{\max} + 1
\end{aligned}
$$

and write

$$V = P \setminus (\mathcal{W} \cup P_{\text{src}})$$

for the walkable cells.

We count clauses rather than literals. The main families admit the following bounds.

- **Initialization.** The agent-initialisation clauses contribute exactly np unit clauses, and the laser-source initialisation contributes exactly $s\tau$ unit clauses.
- **Movement constraints.** Forward consistency contributes at most $n(\tau - 1)p$ clauses. The two uniqueness formulations differ here:

  $$\text{global uniqueness} = n(\tau - 1)\binom{p}{2}$$

  $$\text{local uniqueness} = n(\tau - 1)\sum_{u \in V}\binom{|\text{next}(u)|}{2}$$

  Since $|\text{next}(u)| \leq 5$, the local uniqueness term is bounded by

  $$n(\tau - 1)10\ |V| \leq 10n(\tau - 1)p$$

  and backward consistency contributes at most $n(\tau - 1)p$ further clauses. The blocked-cell clauses contribute

  $$n\tau(|\mathcal{W}| + |P_{\text{src}}|) \leq n\tau p$$

  The collision and conservative hand-off clauses contribute at most

  $$\binom{n}{2}(\tau + 2(\tau - 1))p = \binom{n}{2}(3\tau - 2)p$$

  The exit condition contributes exactly e clauses, and the stay-on-exit condition contributes exactly $ne(\tau - 1)$ clauses.
- **Laser constraints.** The laser-safety clauses contribute at most $ns\tau p$ clauses. Beam propagation contributes at most $3s\tau p$ clauses, because each admissible propagation edge yields either one wall-blocking clause or three equivalence clauses. The beam/laser linking clauses contribute exactly $2s\tau p$ clauses.

Therefore the total number of clauses is polynomial in n, p, s, and τ. More precisely, with the global formulation the dominant term is

$$O(n\tau p^2 + n^2\tau p + ns\tau p)$$

while with the local formulation it is

$$O(n\tau p + n^2\tau p + ns\tau p)$$

Since each clause is generated by a simple bounded computation inside these loops, the reduction itself is computable in polynomial time as well. This justifies the claim that bounded-horizon LLE solvability is polynomial-time reducible to SAT.

4.2.5 Correctness of the Reduction

Proposition. Let L be an LLE level and let $T_{\max}$ be a horizon. For either movement formulation described above, the CNF formula $\Phi(L, T_{\max})$ is satisfiable if and only if there exists a valid joint trajectory of length $T_{\max}$ for L.

Proof sketch. For soundness, assume $\Phi(L, T_{\max})$ is satisfiable. Because initialization, illegal-cell clauses, and the chosen uniqueness formulation constrain every agent-variable family, for each colour c and time t there is exactly one position (x, y) such that $a_{c,x,y,t}$ is true. This defines a trajectory $p_{t(c)}$. Forward and backward consistency enforce legal motion between consecutive steps; the collision clauses enforce both simultaneous separation and the conservative hand-off rule; the laser clauses enforce safety with respect to active beams; the exit clauses enforce the terminal condition. Hence the extracted trajectory is valid.

For completeness, assume a valid joint trajectory of length $T_{\max}$ is given. Set each $a_{c,x,y,t}$ according to whether agent c occupies (x, y) at time t in the trajectory. Set beam variables $b_{c,d,x,y,t}$ and laser variables $l_{c,x,y,t}$ according to the deterministic beam dynamics induced by the same agent positions. Every clause family is then satisfied by construction: initialization matches the start state, movement and collision clauses match the trajectory semantics, beam propagation follows the induced laser dynamics, and the final positions occupy all exits. Therefore $\Phi(L, T_{\max})$ is satisfiable. $\square$

4.3 Cooperation Detection

4.3.1 Strict Laser Semantics

Recall from that the standard encoding allows agent $c \in C$ to occupy a cell where a laser of colour c is active - the agent is immune to its own colour. The **strict** variant removes this immunity: every agent is blocked by every laser, regardless of colour.

Concretely, the strict encoding adds the following constraint for every agent $c \in C$, every position $(x, y) \in P$, and every time step $t \in T$:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} \neg l_{c,x,y,t} \vee \neg a_{c,x,y,t}$$

All other constraints from remain unchanged. We denote the resulting CNF formula by $\Phi_{\text{strict}}(L, T)$ and the corresponding solver by StrictSolver.

4.3.2 Why This Captures Cooperation

Under the LLE mechanics considered in this thesis, the relevant cooperative act is for an agent to position itself on a beam of its own colour, thereby blocking that beam and allowing other agents to traverse cells that would otherwise remain unsafe.

An agent that blocks its own laser receives no direct navigational benefit from doing so: it is already immune to that beam. The beneficiary is a teammate whose path becomes traversable only after the blocking action.

Strict semantics forbids exactly this action. Therefore, if a level is solvable under the standard semantics but unsatisfiable under the strict semantics, every successful standard solution must rely on at least one such blocking step.

4.3.3 Formal Theorem and Proof

Theorem. Let L be an LLE level and T a time horizon. Then L requires cooperation with horizon T if and only if $\Phi(L, T)$ is satisfiable and $\Phi_{\text{strict}}(L, T)$ is unsatisfiable.

Proof.

($\rightarrow$) Assume that L requires cooperation with horizon T. By definition, L is solvable, so $\Phi(L, T)$ is satisfiable. Suppose for contradiction that $\Phi_{\text{strict}}(L, T)$ is also satisfiable. Then there exists a valid strict trajectory σ of length T whose final positions occupy all exit tiles. Because the strict constraints only add restrictions, every strict trajectory is also a valid standard trajectory. But in σ, no agent ever occupies a cell where its own laser is active. Hence σ is a valid standard solution that contains no cooperative laser-blocking act, contradicting the assumption that cooperation is required. Therefore $\Phi_{\text{strict}}(L, T)$ is unsatisfiable. $\square$

($\leftarrow$) Assume that $\Phi(L, T)$ is satisfiable and $\Phi_{\text{strict}}(L, T)$ is unsatisfiable. The first condition implies that L is solvable under the standard semantics. The second condition implies that no valid strict trajectory reaches the exits. Since the only difference between the two semantics is whether an agent may stand on a beam of its own colour, every standard solution must contain at least one time step t, one agent $c \in C$, and one position $(x, y) \in P$ such that both $a_{c,x,y,t}$ and $l_{c,x,y,t}$ hold. Such a step is precisely a same-colour laser-blocking act. Under the LLE mechanics, this act can only help another agent; it does not create a new traversal option for the blocking agent itself. Therefore every successful standard solution requires at least one cooperative act, so L requires cooperation. $\square$

4.3.4 Practical Algorithm

The cooperation detector runs two SAT calls on the same level:

1. Run $\text{Solver}(L, T)$. If the result is UNSAT, the level is unsolvable for horizon T, so it is rejected before cooperation is considered.
2. Run $\text{StrictSolver}(L, T)$. If the result is UNSAT, the level requires cooperation for the same horizon T.

Both calls share the same bounded horizon and differ only in the additional strict constraint clauses. In practice, the detector should always use the same horizon as the standard solvability check. For benchmark levels, this horizon can be chosen from known minimal solution lengths; for generated levels, it is the user-supplied generation parameter T_{max}.

4.4 Level Generators

4.4.1 Design Pattern

All generators follow a common architecture built around three principles:

1. **SAT as oracle**: the solver is not post-hoc; it is embedded in the generation loop. A candidate level is accepted only if the solver confirms the desired property (solvability, cooperation).
2. **Separation of concerns**: level construction and property verification are decoupled. Generators build candidate levels using domain-specific heuristics; the solver decides acceptance.
3. **Extensibility**: every generator extends `BaseGenerator` and is registered via the `@register_generator` decorator, making it available to the CLI without modifying core code.

In implementation terms, each generator repeatedly performs the following loop: sample or construct a candidate layout, reject it if it violates generator-specific structural constraints, build an `lle.World`, and finally run the appropriate SAT-based acceptance test. Solvable generators stop after the first candidate certified satisfiable within the target horizon, while cooperative generators add the strict-semantics counterfactual test and, optionally, a cooperation-profile filter.

4.4.2 Generation Targets

Viewed through the solvability and cooperation definitions of , the generator family targets the three level categories shown in Figure Figure 2. Solvable generators accept levels in categories (b) and (c), cooperative generators accept only levels in category (c), and unsolvable levels in category (a) are always rejected.

(a) Unsolvable
rejected by all generators

(b) Solvable, no cooperation
accepted only by solvable generators

(c) Solvable and cooperative
target of cooperative generators

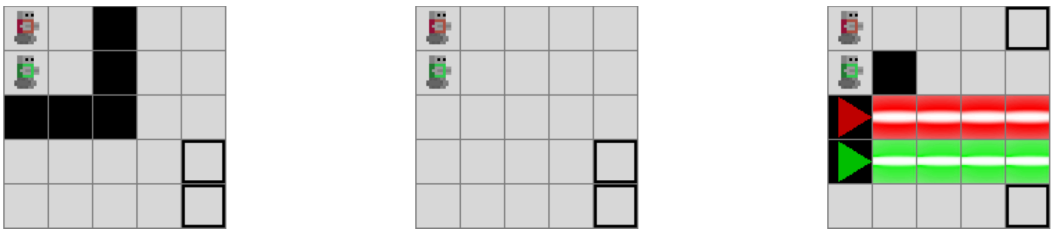

Figure 2: Target level categories for the generator family.

4.4.3 Random Solvable Generator

The random solvable generator is the baseline member of the family. It samples pairwise-distinct positions for agent starts, exits, walls, and laser sources uniformly over the grid, assigns a random direction to each source, and submits the resulting world to the solver. A candidate is accepted only if it is satisfiable within the requested horizon $T_{\max}$; when a lower bound $T_{\min}$ is provided, the generator also requires the candidate to be unsatisfiable for $T_{\min} - 1$, thereby selecting levels that fall inside a difficulty window.

This generator is deliberately simple. Its main value is methodological: it gives an unbiased sampling baseline against which more structured generators can be compared. Its main weakness is rejection rate. As the grid grows and the number of interacting entities increases, purely random layouts quickly become dominated by unsolvable or trivial instances.

4.4.4 Constrained Random Solvable Generator

A structured variant that biases generation toward solvable configurations before any SAT call is made. Relative to the random solvable generator, it rejects candidates that are already geometrically degenerate, for example when a laser points outside the grid immediately, when a laser would have zero beam length, or when an exit lies on an unavoidable beam segment.

These filters do not themselves prove solvability, but they remove a large class of obviously bad candidates before invoking the solver. The generator therefore remains sound with respect to the formal solvability guarantee, while typically spending less time on layouts that fail for purely local geometric reasons.

4.4.5 Random Cooperative Generator

The random cooperative generator extends the random solvable generator with a second SAT test based on the strict semantics of . A candidate is accepted only if it is satisfiable under the standard encoding and unsatisfiable under the strict encoding. This guarantees that every accepted level structurally requires inter-agent coordination.

The current implementation augments this binary guarantee with a **cooperation profile analyzer**. The binary detector remains the formal guarantee used throughout the thesis: a level is cooperative if and only if it is satisfiable under the standard semantics and unsatisfiable under the strict semantics. The analyzer adds a second layer whose purpose is to distinguish **which kind** of cooperation the accepted level exhibits.

Given a cooperative level, we first extract a valid joint plan from the standard SAT model. We then run selective counterfactual checks in which one agent at a time loses the ability to block beams of its own colour. If the level becomes unsatisfiable under that selective restriction, the agent is identified as a necessary helper. By combining these counterfactual checks with the helping actions observed in the extracted plan, we build a directed dependency graph between agents.

This dependency graph is used as a generation target. In the present implementation, the generator can recognise and filter levels according to the following profile families:

- **cooperative**: binary cooperation is required, regardless of finer structure;
- **asymmetric**: at least one one-way helping relation is present;
- **mutual**: two agents depend on each other;
- **chain**: dependencies form a directed chain without branching;
- **distributed**: one agent depends on multiple distinct helpers;
- **fully coupled**: all agents belong to a single strongly connected dependency component.

The important methodological point is that profile control is layered on top of the existing formal machinery. The SAT encodings still certify solvability and binary cooperation. The profile analyzer uses those certified levels as input and acts as a classification and filtering layer for the generator.

4.4.6 Constrained Random Cooperative Generator

The constrained random cooperative generator combines the geometric filters of the constrained solvable generator with the binary cooperation test and optional profile filter of the random cooperative generator. In other words, it first avoids immediately degenerate geometries, then requires the surviving candidates to satisfy the same solver-based cooperation criterion.

This generator therefore targets the same formally certified output class as the random cooperative generator, but with a sampling distribution biased away from trivial failures. It is useful when the goal is not only to obtain cooperative levels, but to obtain them with fewer discarded samples.

4.4.7 Constructive Solvable Generator

The constructive solvable generator replaces blind sampling with a partial-by-construction layout. It reserves one disjoint horizontal or vertical lane per agent, places each start at one end of its lane and the corresponding exit at the other end, and only samples walls and lasers outside the reserved traversable lanes. Additional lasers are accepted only if their beam segments avoid the reserved cells. The solver still acts as the final verifier, but the sampling process is strongly biased toward jointly solvable instances.

4.4.8 Constructive Cooperative Generator

The constructive cooperative generator further specialises the constructive idea by planting a deliberate dependency pattern. One laser is placed so that a helper agent must block a beam of its own colour before a beneficiary lane becomes traversable. The resulting candidate is then verified with the standard and strict SAT encodings, and can optionally be filtered by cooperation profile. This generator is useful when one wants deliberately cooperative instances rather than merely sampling for cooperation and hoping to find it by rejection.

4.4.9 Summary

Generator	Construction Bias	Solvable	Cooperative
Random Solvable	Uniform random sampling	Yes (SAT check)	No
Constrained Random Solvable	Random + geometric rejection	Yes (SAT check)	No
Random Cooperative	Uniform random sampling	Yes (SAT check)	Yes (strict UNSAT)
Constrained Random Cooperative	Random + geometric rejection	Yes (SAT check)	Yes (strict UNSAT)
Constructive Solvable	Reserved agent lanes	Yes (SAT check)	No
Constructive Cooperative	Reserved lanes + planted dependency	Yes (SAT check)	Yes (strict UNSAT)

Table 1: Overview of the implemented generators and their guaranteed properties.

4.5 Benchmarking Setup

The first benchmark implemented in this project isolates the SAT solver rather than the generators. Its goal is to compare the two movement formulations used in the encoding: the **local** formulation, which combines neighbourhood-based exclusivity with backward consistency, and the **global** formulation, which enforces uniqueness by pairwise exclusion over the whole grid.

4.5.1 Metrics

For each run, the benchmarking code records:

- the total number of clauses in the generated CNF;
- the clause count contributed by each major constraint family;
- the CNF generation time;
- the SAT solving time;
- the total time obtained by summing generation and solving.

The implementation also stores per-constraint method profiles for the movement constraint, which allows us to inspect how much of the final CNF is attributable specifically to the local or global uniqueness mechanism.

4.5.2 Solver and Level Sets

All benchmark runs use the same SAT backend as the main solver implementation, namely `Minisat22` through the PySAT interface. By default, the benchmarking script can evaluate the six hand-crafted LLE levels listed in `levels.py`, each paired with a horizon known to be sufficient for solvability. It can also benchmark custom levels constructed programmatically with `WorldBuilder`.

The experiment reported in Chapter 5 uses four representative levels: three synthetic instances of increasing size and one original LLE level. This combination exposes both scaling behaviour and the behaviour of the solver on a realistic cooperative puzzle.

4.5.3 Protocol

For each level and each movement formulation, the benchmark performs one profiled run to extract the exact clause counts and the full constraint breakdown. It then repeats the same solver invocation for 100 runs, each time on a fresh copy of the world, and reports the mean and standard deviation of generation time and solve time. Using fresh world copies avoids benchmark contamination by mutable environment state.

No timeout or parallel speedup is introduced in this protocol. The measurements should therefore be read as direct comparisons between the two SAT formulations on the same machine, rather than as hardware-independent absolute performance claims.

4.5.4 Outputs

The benchmarking pipeline produces a console summary table, a JSON file containing the raw measurements, and a set of plots for clause counts, per-constraint clause breakdowns, and timing statistics. Chapter 5 draws on these outputs to interpret the trade-off between the two movement formulations.

Chapter 5

Experiments

The goal of this first experimental evaluation is to compare two alternative formulations of the agent uniqueness constraint introduced in , and to measure their effect on the size of the CNF formula and on the time required by the SAT solver across levels of increasing complexity.

5.1 Experimental Setup

This experiment follows the protocol of . All runs use the `Minisat22` backend through PySAT. For each level and each movement formulation, the benchmark records one exact clause profile and then repeats the solve 100 times to obtain mean generation and solve times with standard-deviation error bars.

The analytical comparison is driven by the uniqueness constraint itself. The **global** formulation adds a clause for every unordered pair of positions, yielding a quadratic contribution in $|P|$. The **local** formulation restricts exclusivity to neighbourhood-sized successor sets and supplements it with backward consistency, yielding a linear contribution in $|P|$ because $|\mathrm{next}(x, y)| \leq 5$. The blocked-cell clauses introduced in are shared by both formulations and therefore do not affect the comparison between them.

More precisely, if we isolate the **method-specific** contribution of the uniqueness mechanism, then for one agent and one time step we obtain:

$$\text{global uniqueness} = \binom{|P|}{2} = \frac{(|P| - 1)\ |P|}{2}$$

$$\text{local uniqueness} + \text{backward consistency} = \sum_{u \in V} \binom{|\mathrm{next}(u)|}{2} + |V|$$

where V denotes the set of walkable cells. Since $|\mathrm{next}(u)| \leq 5$ for every $u \in V$, this local term satisfies

$$\sum_{u \in V} \binom{|\mathrm{next}(u)|}{2} + |V| \leq \left(\binom{5}{2} + 1 \right)\ |V| \leq 11\ |P|$$

This yields the theoretical crossover used to interpret the plots: the global contribution is smaller on very small instances, but once

$$\binom{|P|}{2} > 11\ |P|$$

the local contribution becomes smaller. The upper-bound crossover occurs at $|P| = 23$, where the two expressions are equal.

5.1.1 Test Levels

Four levels of increasing size and complexity were used:

3x3	**5x5**	**8x8**	**LLE Level 6**
2 agents, 1 laser	3 agents, 2 lasers	4 agents, 3 lasers	4 agents, 3 lasers

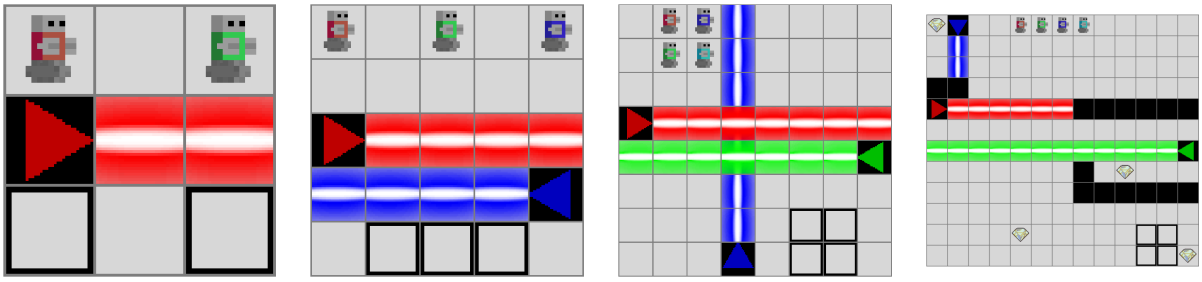

Figure 3: The four test levels used for benchmarking, ordered by increasing complexity.

The first three levels were constructed specifically to test scaling behaviour across grid size and agent count. The fourth is LLE Level 6, one of the hand-crafted levels shipped with the LLE benchmark, included to assess solver performance on a realistic cooperative level.

5.2 Results

5.2.1 Clause Count

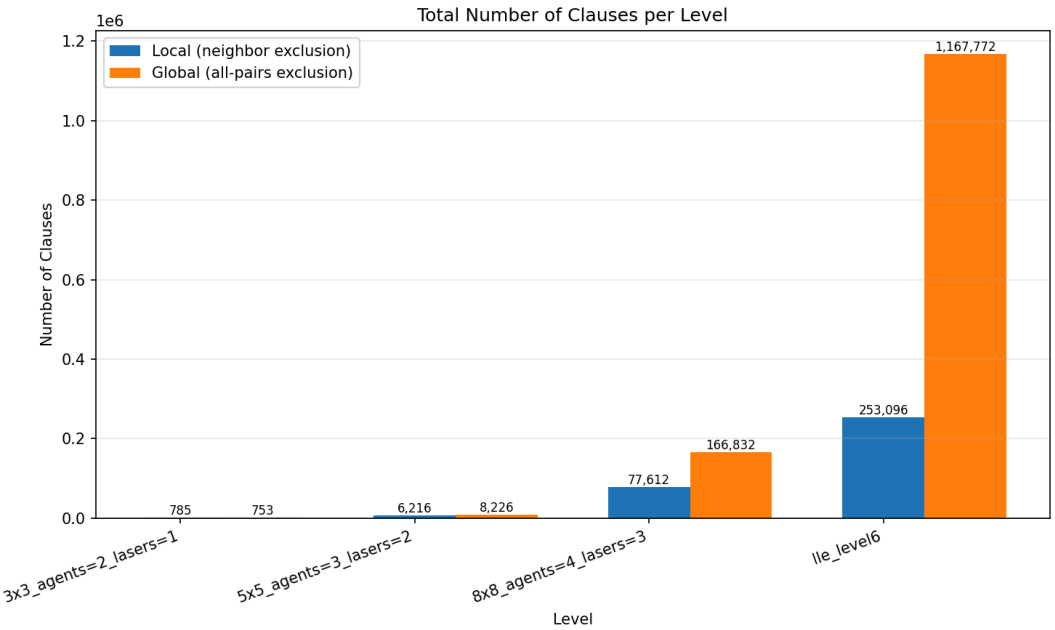

Figure 4: Number of clauses generated by the global and local uniqueness formulations across the four test levels.

The clause-count results closely match the theoretical expectation. On the smallest 3×3 level, the global formulation is slightly more compact than the local one (753 clauses versus 785), which is consistent with the fact that the instance lies below the expected crossover regime. From the 5×5 level onward the trend reverses: the local formulation generates 6,216 clauses versus 8,226 for the global one, then 77,612 versus 166,832 on the 8×8 level, and finally 253,096 versus 1,167,772 on LLE Level 6. The gap widens rapidly as the number of walkable cells and interacting agents increases, which is precisely the behaviour predicted by the linear versus quadratic growth of the two formulations.

5.2.2 Solving Time

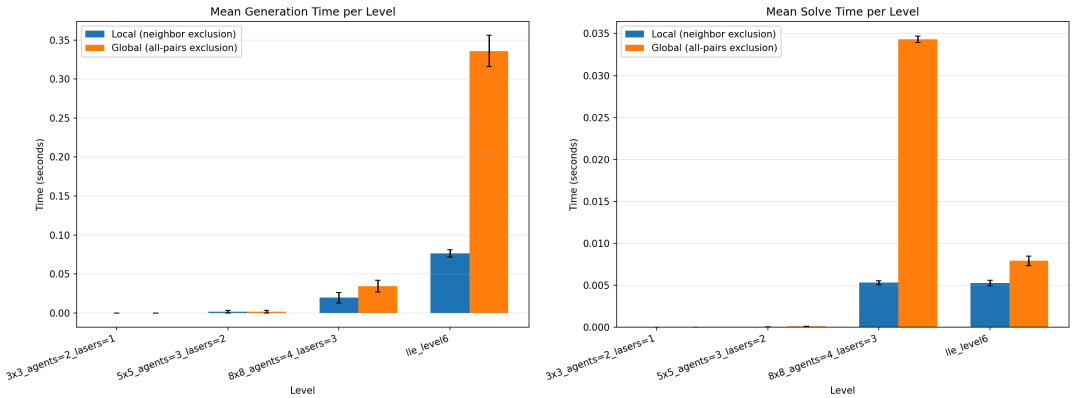

Figure 5: SAT solver time (seconds) for the global and local uniqueness formulations across the four test levels.

The timing results show the same qualitative picture. On the smallest synthetic levels both methods solve essentially instantaneously, and the difference is negligible. Once the instances become moderately large, however, the local formulation is consistently faster both to generate and to solve. The contrast is especially visible on the 8×8 level, where the global formulation has the largest solve-time penalty, and on LLE Level 6, where the cost of generating the much larger global CNF dominates total runtime. In other words, the clause-count reduction is not merely a syntactic improvement; it translates into a measurable performance gain.

5.3 Discussion

The experiment supports a clear design choice. The global formulation can be marginally attractive on very small grids because it introduces fewer clauses there, but this advantage disappears almost immediately. For the levels of practical interest in this thesis, the local formulation is both more compact and faster. We therefore treat the local encoding as the preferred default in the implementation.

The broader experimental agenda remains open. In particular, cooperation-detection cost, generator acceptance rates, and structural diversity of accepted levels should be studied in a dedicated second experimental phase. The current benchmark nonetheless establishes an important baseline: the movement formulation chosen inside the SAT encoding has a first-order effect on both CNF size and runtime, and that effect is already substantial on realistic LLE instances.

Conclusion

6.1 Summary

This thesis developed a SAT-based framework for generating LLE levels with formal guarantees. We first reduced bounded-horizon LLE solvability to satisfiability of a CNF formula, then defined a strict laser semantics that turns cooperation detection into a second SAT query on the same level. On top of these decision procedures, we implemented a family of random, constrained, and constructive generators whose outputs are certified by the solver before acceptance.

The first experimental evaluation compared two movement formulations inside the SAT encoding. The results showed that the local formulation is the better default for the levels studied here: once the instances move beyond the smallest toy case, it produces substantially smaller CNF formulas and lower runtimes than the global all-pairs formulation.

6.2 Limitations

The current framework remains deliberately scoped. The encoding is tailored to the subset of LLE mechanics needed for solvability and cooperation analysis, so features such as gems and void tiles are abstracted away. The guarantees are also horizon-bounded: the solver decides whether a level is solvable within a fixed $T_{\max}$, not whether it is solvable under an unbounded notion of play. Finally, the empirical results reported here cover only the movement-formulation comparison; they do not yet constitute a full evaluation of generator acceptance rates, diversity, or downstream learnability.

6.3 Future Work

Several natural extensions follow from these limitations.

- Extend the experimental section with generator-focused studies: acceptance rates, cooperation profile frequencies, and diversity measures across parameter regimes.
- Generalise the encoding to richer LLE mechanics or to other cooperative grid environments, while preserving the same solver-in-the-loop methodology.
- Investigate whether learnability can be linked to structural properties of the generated levels, for example by combining the current formal guarantees with curriculum design or MARL training experiments.
- Refine the cooperation-profile analysis so that it becomes not only a filtering mechanism for generators, but also an evaluation tool for the qualitative structure of cooperative tasks.

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